



Linux File Permissions

✓ Multiple Choice Questions (MCQs)

Instructions: Answer all questions. Each question carries **1 mark**.

Form Link: <https://forms.gle/9DE8fwJf1GT2PHu69>

1 What does the first character in `ls -l` output represent?

- A) Owner
- B) Permission type
- C) File type
- D) File size

✓ **Answer: C**

2 What does the permission string `-rw-r--r--` mean for the owner?

- A) Read only
- B) Read & Write
- C) Read, Write & Execute
- D) Execute only

✓ **Answer: B**

3 What does execute (x) permission mean for a directory?

- A) Modify files
- B) Delete files
- C) Enter the directory
- D) View file content

✓ **Answer: C**

4 Which numeric value represents **r-x**?

- A) 4
- B) 5
- C) 6
- D) 7

 **Answer: B**

5 What does **chmod 755 file.txt** mean?

- A) Everyone full access
- B) Owner full, others read & execute
- C) Only owner can access
- D) No one can execute

 **Answer: B**

6 Which command changes file ownership?

- A) chmod
- B) chgrp
- C) chown
- D) umask

 **Answer: C**

7 Which permission is required to delete a file?

- A) Write on file
- B) Execute on file
- C) Write & Execute on directory
- D) Read on directory

 **Answer: C**

8 What does SUID affect?

- A) Real UID
- B) Effective UID
- C) Real GID
- D) File size

 **Answer: B**

9 Which numeric value enables SUID?

- A) 1
- B) 2
- C) 4
- D) 7

 **Answer: C**

10 Which of the following shows SUID enabled?

- A) -rwxr-xr-x
- B) -rwsr-xr-x
- C) drwxr-xr-x
- D) -rw-r--r--

 **Answer: B**

11 What does SGID do on directories?

- A) Makes directory executable
- B) Changes file owner
- C) New files inherit group
- D) Deletes files automatically

 **Answer: C**

12 Which numeric value enables SGID?

- A) 1
- B) 2

- C) 4
- D) 6

 **Answer: B**

13) Sticky bit is mainly used on?

- A) Regular files
- B) Executables
- C) Directories
- D) Devices

 **Answer: C**

14) Which command sets sticky bit on a directory?

- A) `chmod 2777 dir`
- B) `chmod 4777 dir`
- C) `chmod 1777 dir`
- D) `chmod 0777 dir`

 **Answer: C**

15) What is the default permission of a newly created file before umask?

- A) 777
- B) 666
- C) 755
- D) 644

 **Answer: B**

16) If umask is 0022, what will be the permission of a new file?

- A) 777
- B) 666
- C) 644
- D) 755

✓ Answer: C

17 What does `chmod 600 file.txt` mean?

- A) Everyone full access
- B) Owner read/write only
- C) Owner execute only
- D) Group read/write

✓ Answer: B

18 Which command changes only group ownership?

- A) `chown`
- B) `chmod`
- C) `chgrp`
- D) `umask`

✓ Answer: C

19 What does uppercase **T** in permission string indicate?

- A) No sticky bit
- B) Sticky bit + execute
- C) Sticky bit without execute
- D) Execute only

✓ Answer: C

20 What does `umask` do?

- A) Adds permissions
- B) Removes default permissions
- C) Changes owner
- D) Enables SUID

✓ Answer: B

